AAMOD VARMA

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EDUCATION

Georgia Institute of Technology Atlanta, GA
B.S. Computer Science and B.S. Mathematics, ***GPA: 4.0/4.0*** *Expected May 2027*
Coursework: Machine Learning, Deep Learning, Data Structures and Algorithms, Analysis, Operating Systems, Compilers
Honors: 1st Place, Georgia Tech Hacklytics Traversaal AI Challenge (2024)

PUBLICATIONS

- Jia, S., **Varma, A.**, Manivannan, P., Chayapathy, D., & Fung, V. **"Benchmarking Text Representations for Crystal Structure Generation with Large Language Models."** *ICLR 2025 AI4Mat Workshop, full paper*.
- Candocia, A., Inan, O. M., Agarwal, R., **Varma, A.**, & Davenport, M. A. **"Scalable Continual Learning: Adaptive MoEs for Expanding Task Sets."** *ICLR 2025 Workshop on Sparsity in LLMs (SLLM)*, short paper.

EXPERIENCE

Fung Group, Georgia Tech — advised by Prof. Victor Fung Oct 2023 - Present
Undergraduate Research Assistant *Atlanta, GA*

- Authored manuscript showing synthetic crystals replace DFT in diffusion pretraining, achieving an average **31%** MAE reduction across **7** Matbench tasks and matching DFT pretraining, by evaluating multi-source synthetic datasets.
- Built an equilibrium structure pipeline (perturb-and-relax, MatterGen sampling, protostructure enumeration), eliminating costly DFT relaxations by using MACE ML interatomic potentials to relax **165K+** synthetic structures at scale.
- Demonstrated joint data-compute scaling for synthetic pretraining, matching 20K DFT pretraining at **75K** and surpassing it on **7 of 8** Matbench tasks at **110K–165K**, by scaling protostructures.
- Ablated equivariant diffusion objectives on the Orb-v2 backbone, cutting jdft2d MAE by **37%** and beating coordinate denoising across all tasks, by formulating continuous-time VP and sub-VP coordinate diffusion SDEs.

Research Group, Georgia Tech — advised by Prof. Mark Davenport Jul 2024 - Present
Undergraduate Research Assistant *Atlanta, GA*

- Co-developed global convergence theory and led experiments for gradient EM on MoEs, validating the $m \log m$ width scaling transition and showing unconstrained models prune to $k \approx m$, by sweeping widths ($n \leq 100$) and router penalties.
- Characterized capacity-driven task interference for a DeepMath 2025 poster, showing width cuts test error by **45%** (**3.5 to 1.9**) while depth drives gradient conflict to **−0.92**, by evaluating pairwise gradient similarity across MNIST sweeps.
- Developed adaptive MoE routing for expanding continual-learning task sets; co-authored an ICLR 2025 SLLM Workshop paper.
- Designed covariance-aware routing for Gaussian tasks using Mahalanobis distance and log-determinant scoring; benchmarked across task streams on synthetic subspace data, MNIST, and CIFAR-10/100.

PRGX Jan 2025 - Jul 2025
AI Intern *Atlanta, GA*

- Architected a French-to-English contract translation pipeline, preserving PDF layout fidelity, by aligning OCR text to source bounding boxes using Sentence-Transformers and spaCy.
- Accelerated table extraction by **90%** and raised TEDS from **0.50 to 0.80** by fine-tuning Qwen-VL and PaddleOCR.
- Enhanced contract key-field extraction, improving parsing accuracy by **80%**, by fine-tuning BERT and Mistral NER models.

School of Mathematics, Georgia Tech Feb 2024 - Jan 2025
Software Developer *Atlanta, GA*

- Built a Python/OpenAI API pipeline that generated and formatted WeBWorK, Canvas, and question-bank content for multivariable calculus courses serving 380+ students.

PROJECTS

Structured-Mask Diffusion for Image Restoration May 2025 - Jul 2025

- Trained **200K**-step mask-conditioned U-Net DDPMs in PyTorch with edge and semantic priors on FFHQ using H100/H200 GPUs, mixed precision, and EMA; reduced FID by **58%** (**48.0 to 20.3**).
- Ablated mask conditioning, mask-weighted loss, and adaptive noise; identified conditioning as the only robust gain and traced adaptive-noise regressions to a violated Gaussian forward-process assumption.

TraverGo - Georgia Tech Hacklytics Winner Feb 2024 - Mar 2024

- Built a hotel-search platform combining semantic matching using Qdrant, review sentiment analysis, and a real-time Q&A chatbot; placed **1st** in the Traversaal AI Challenge among **1,000** hackathon participants.

SKILLS

Programming: Python, C/C++, Java, SQL, Bash, R; Linux, Git, LaTeX
ML Frameworks: PyTorch, NumPy, Pandas, Sentence-Transformers, spaCy, PaddleOCR, MatterTune, MACE
Research Areas: Diffusion and score-based models, Scientific machine learning, Equivariant GNNs, Continual learning and MOEs, Optimization/Convergence Analysis, Vision-language modeling, GPU/mixed precision training.